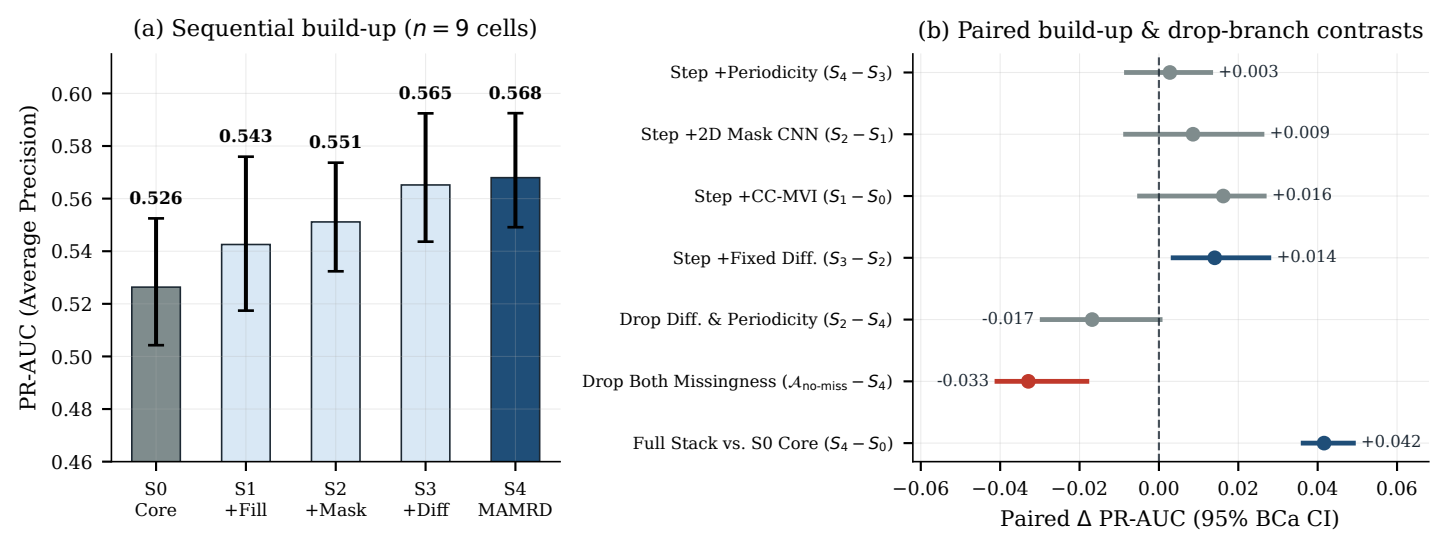
(a) Sequential build-up ($n = 9$ cells)
PR-AUC (Average Precision)
0.60
0.58
0.56
0.54
0.52
0.50
0.48
0.46
0.526
0.543
0.551
0.565
0.568
S0 Core
S1 +Fill
S2 +Mask
S3 +Diff
S4 MAMRD
(b) Paired build-up & drop-branch contrasts
Step +Periodicity ($S_4 - S_3$)
+0.003
Step +2D Mask CNN ($S_2 - S_1$)
+0.009
Step +CC-MVI ($S_1 - S_0$)
+0.016
Step +Fixed Diff. ($S_3 - S_2$)
+0.014
Drop Diff. & Periodicity ($S_2 - S_4$)
-0.017
Drop Both Missingness ($A_{\text{no-miss}} - S_4$)
-0.033
Full Stack vs. S0 Core ($S_4 - S_0$)
+0.042
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
Paired Δ PR-AUC (95% BCa CI)